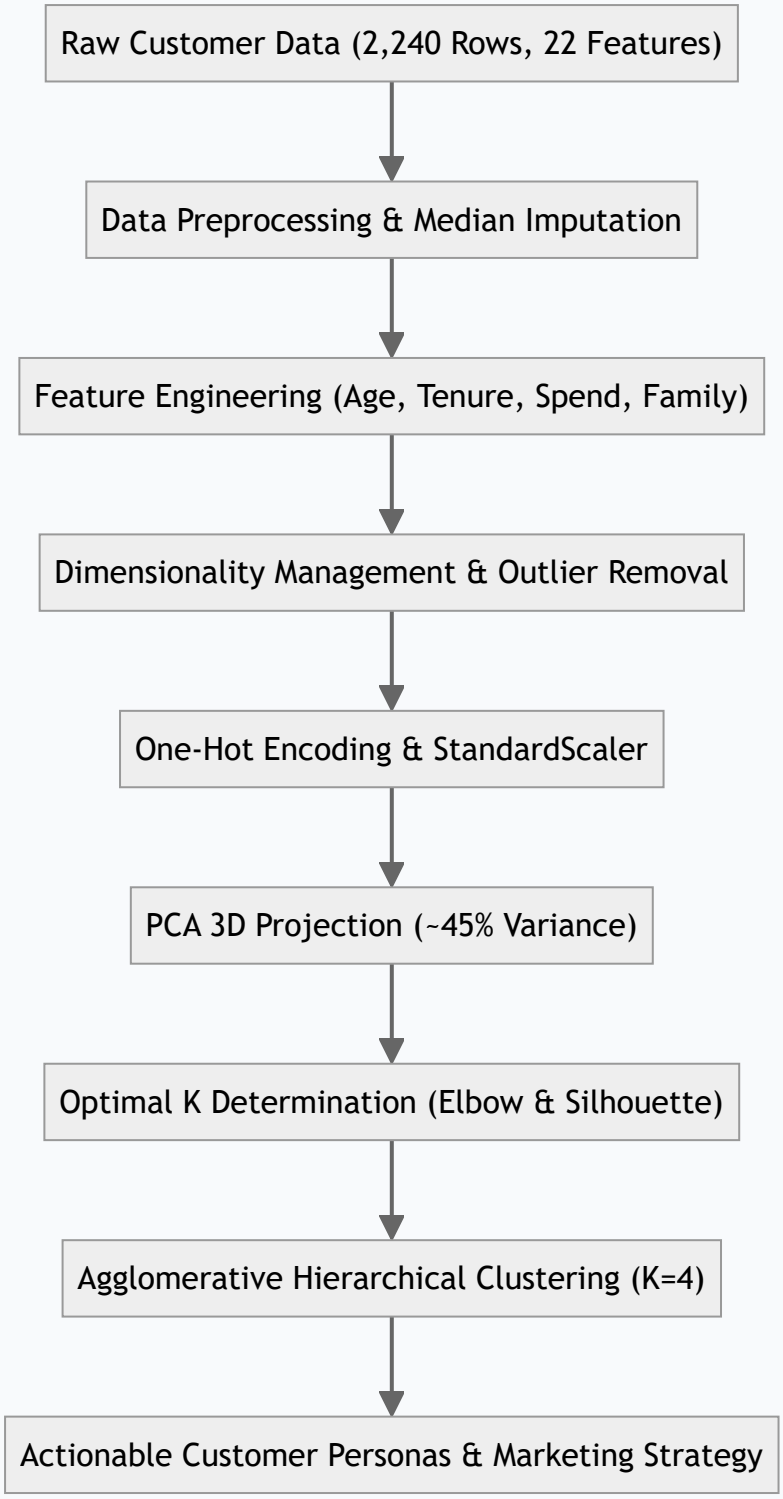
Submodule 33.01: Project Introduction & Business Objectives (Hinglish)

Topic Overview & Business Context

E-commerce aur modern retail platforms ke andar customer segmentation **Unsupervised Machine Learning** ka sabse popular aur commercially impactful use case hai. Agar ek company apne sabhi customers ko ek jaisa treat karegi aur generic marketing strategies ("spray-and-pray") chalayegi, toh marketing budget waste hota hai aur high-value customers retain nahi ho pate.

Is minor project mein hum **SmartCart Clustering System** build kar rahe hain jo ek international e-commerce company ke 2,240 customers ke transaction records (22 attributes) par based hai.



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1. Business Problem aur Objectives

- **Current Problem**: SmartCart generic marketing aur mass engagement strategies use kar raha hai.
- **Nuksan**:
    - Company high-value loyal customers ko identify nahi kar pa rahi hai.

    - Inefficient marketing spend: Low-budget shoppers aur premium spenders ko same campaigns bheje ja rahe hain.
- **Machine Learning Solution**: Unsupervised clustering algorithms use karke historical transaction data se hidden patterns discover karna aur customers ko meaningful segments mein group karna.
- **Goal**: Har segment ke liye personalized marketing, discount targeting aur retention strategies design karna.

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2. Dataset Architecture aur Attributes

Dataset (`smartcart_customers.csv`) ke 22 attributes ko 4 broad categories mein classify kiya gaya hai:

Category	Attributes	Significance
Demographics	`ID` , `Year_Birth` , `Education` , `Marital_Status` , `Income` , `Kidhome` , `Teenhome`	Customer ki age, education, household income aur ghar mein bacchon ki sankhya.
Product Expenditures	`MntWines` , `MntFruits` , `MntMeatProducts` , `MntFishProducts` , `MntSweetProducts` , `MntGoldProds`	Wine, fruit, meat, fish, sweets aur gold par pichle 2 saal mein kitna spend kiya.
Purchasing Channels	`NumDealsPurchases` , `NumWebPurchases` , `NumCatalogPurchases` , `NumStorePurchases`	Discounts, website, catalog aur physical stores se purchases ka count.
Engagement & Recency	`Dt_Customer` , `Recency` , `NumWebVisitsMonth` , `Complain` , `Response`	Joining date, last purchase ke days, monthly web visits, complaints aur campaign response.

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3. Data Ingestion aur Initial Inspection

Official project assets yahan available hain:

- smartcart.ipynb
- smartcart_customers.csv

Python Implementation (Notebook Cells 00 - 04)

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Dataset load karna
df = pd.read_csv("smartcart_customers.csv")

# Pehle 5 rows inspect karna
print(df.head())

# Dimensions check karna
print("Shape:", df.shape) # Output: (2240, 22)

# Null values check karna
print(df.isnull().sum())
```

Initial Observations:

1. Dataset mein **2,240 rows** aur **22 columns** hain.
2. Sirf `Income` column mein **24 missing values** hain, baki sabhi 21 columns completely clean hain.
3. Koi target label (y) nahi hai, yeh pure unsupervised learning problem hai.